Ideal Gases

Describing Gases

Atoms and molecules can't be counted one by one since they are tiny, so we count them in units called "Moles" - just like a "dozen".

Equations

① $N = n \cdot N_A$ ② $m = n \cdot M$

- M : Molar mass
- n : amount in substance (unit : mol)
- m : total mass
- N : total number of molecules

Remember that both equations can be rearranged depending on desired value

The Mole

One mole (n = 1) = the amount of carbon-12 atoms in a sample of 12 grams [6.02×10^{23}]

- Unit : mol

Avogadros constant (N_A) : Number of particles per unit amount of substance.

- Unit : mol^{-1}
- Value = 6.02×10^{23}

Molar mass (M) : Mass of 1 mole

- Unit : $kg\ mol^{-1}$
- Value = no. protons + neutrons

E.g
- 1 mole of He = 6.02×10^{23} helium atoms
- 1 mole of O_2 = 6.02×10^{23} oxygen molecules

Brownian Motion

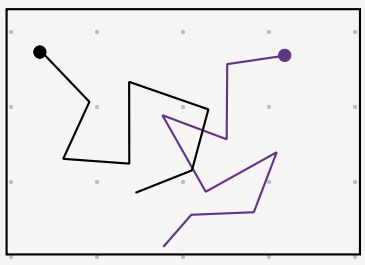
A gas is made up of a huge number of particles moving rapidly and randomly in all directions

Evidence

Under the microscope we can see dust/smoke particles (much larger) moving countinuosly and randomly, this proves that :

① Air molecules are small enough to be invisible, yet much faster to transfer momentum.

② Air molecules are in constant, random motion.



Gases Exerting Pressure

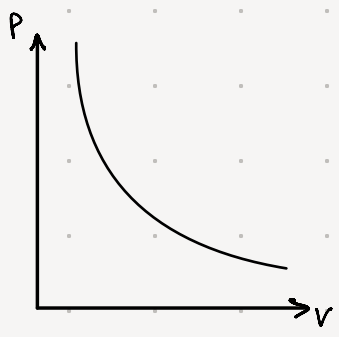
Pressure : Avg. force exerted by gas molecules per unit area on walls of container

Mechanism :
- Molecules collide with walls of container
- Momentum of molecule changes during collision
- This change is caused by force on molecule
- So there must be a force on wall (Newtons 3rd law)
- Many molecules exert force across area → pressure on walls of container

Gas laws

Boyle's law

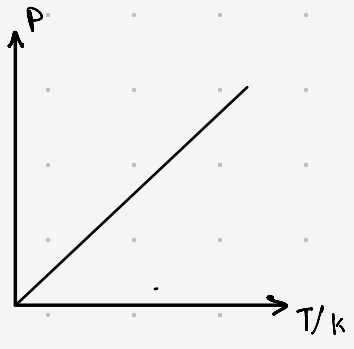
- Relation : $p \propto \frac{1}{V}$, $P_1V_1 = P_2V_2$ (p : Pressure, V : Volume)
- Condition : Constant Tempreture



- Why ?
  - Volume decreases
  - The molecules have the same avg speed → same avg force per collision
  - However, num. of molecules per unit volume increases
  - More collision per unit time → greater avg force → pressure increases

Gay lussa's law

- Relation : $P \propto T$, $\frac{P_1}{T_1} = \frac{P_2}{T_2}$ (T : Tempreture (K))
- Condition : Constant Pressure

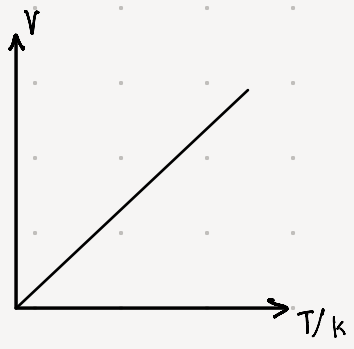


- Why ?
  - Tempreture increases
  - Avg Kenetic energy increases → Avg Velocity increases
  - Gas molecules collide frequently + greater avg force per collision
  - Pressure increases

Gas laws [Continued]

Charle's law

- Relation: $V \propto T$, $\frac{V_1}{T_1} = \frac{V_2}{T_2}$
- Condition: Constant Pressure



- Why?
  - Tempreture increases
  - Avg Kenetic energy increases → avg velocity increases
  - Gas molecules collide frequently + greater avg force per collision
  - Pressure inc., the piston is free to move so volume increases until $P_{in} = P_{atm}$

Using those 3 laws we can deduce that for a fixed amount of gas: $\frac{pV}{T} = \text{Const}$

Ideal Gas Model

Ideal gas: gas that obeys $pV \propto T$, where T is thermodynamic tempreture (for all values of p, V and T)

Equations of state for an ideal gas:

① $pV = nRT$ ② $pV = NkT$

Where:

- P: Pressure (Pa)
- V: Volume (m^3)
- T: Thermodynamic tempreture (K)
- R: Universal gas constant (8.31)
- k: Boltzman Constant

$k = \frac{R}{N_A}$

Kinetic Energy Formula

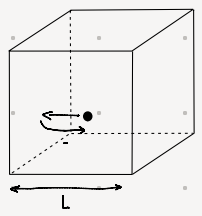
- $p = \frac{1}{3}\rho\langle c^2\rangle \rightarrow pV = \frac{1}{3}Nm\langle c^2\rangle$
- $pV = NkT = \frac{1}{3}Nm\langle c^2\rangle$
- $m\langle c^2\rangle = 3kT \rightarrow \frac{1}{2}m\langle c^2\rangle = \frac{3}{2}kT$
- $\frac{3}{2}kT = \langle K.E.\rangle$ → avg translational K.E. for one molecule

Kinetic Theory

The kinetic model of gases uses microscopic properties to explain macroscopic properties. The kinetic theory of gases is based on the following assumptions:

- gas molecules are in constant random motion
- gas molecules have negligible volume compared with volume of gas
- collisions are instantaneous
- collisions are elastic
- no intermolecular forces – no potential energy

Imagine a single molecule in cube of length "L", assume that it only moves in the x-axis with speed c_x.



- momentum = mc
- change in momentum = $m(c_x - -c_x) = 2mc_x$
- the time between collision on wall = $\frac{2L}{c_x}$
- So, avg force = $\frac{2mc_x}{(\frac{2L}{c_x})} = \frac{mc_x^2}{L}$ → avg pressure = $\frac{mc_x^2}{V}$

Lets generalize for multiple molecules

- For N molecules, pressure = $\frac{\overline{Nm}c_x^2}{V}$

total mass $\frac{\overline{Nm}}{V}$ = density = ρ

→ Pressure = $p = \rho c_x^2$

Lets move to 3D

- molecules move randomly in 3D (x, y, z) not one, and by symmetry $c_x^2 = c_y^2 = c_z^2$
- total speed $c = \sqrt{c_x^2 + c_y^2 + c_z^2}$ → $c^2 = 3c_x^2$
- Speed is not the same for all molecules, so we use averge speed → $\langle c_x^2\rangle = \frac{1}{3}\langle c^2\rangle$
- Finally, $p = \frac{1}{3}\rho\langle c^2\rangle$ and $pV = \frac{1}{3}Nm\langle c^2\rangle$

Mean square speed ($\langle c^2\rangle$): Avg of square of speeds $\frac{c_1^2 + c_2^2 + \dots}{n}$

Root mean square speed (c_{rms}): Square-root of mean square speed

$c_{rms} = \sqrt{\langle c^2\rangle}$